

Applying Newton's Laws

Building Models to Describe our World

Motion, Derivations and Modelling Situations



Outline

Types of Motion

- Linear motion

- Constant net force

- Varying net force

 - Varying with time

 - Varying with position

- Breaking up motion

- Force on a spring

Mathematical Applications

- Derrivation for force varying with position

- Differential equations

- Force varying with velocity

- Example: linear drag force

Building Models

- Applying Newton's Laws

- Circular motion with Newton's laws

- Two masses on accelerating turntable

- Uniform circular motion

- Banked curves

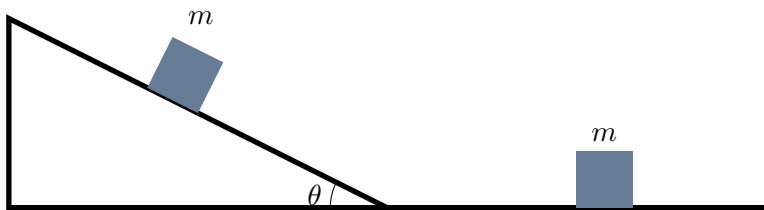
- Vertical Circle

Types of Motion

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Linear motion

- We can loosely define “linear motion” as motion along a straight line, where the velocity does not change direction.
- The net force (and acceleration) is co-linear with the velocity.
- We can break up many types of motion into segments where the motion is linear along the segments.



A block slides down an incline and then a flat section. Both segments are linear motion.

Constant net force

- When the **sum of the forces** (aka the net force) on an object is constant (in magnitude and direction), and the mass of the object is constant, the acceleration of that object is constant:

$$\sum \vec{F} = \vec{F}^{net} = m\vec{a}$$

- When the acceleration is constant:

$$\vec{v}(t) = \vec{v}_0 + \vec{a}t = \vec{v}_0 + \frac{\vec{F}^{net}}{m}t$$

$$\vec{r}(t) = \vec{r}_0 + \vec{v}_0t + \frac{1}{2}\vec{a}t^2 = \vec{r}_0 + \vec{v}_0t + \frac{1}{2}\frac{\vec{F}^{net}}{m}t^2$$

- This is a vector equation, so true by component, for example, the x component:

$$v_x(t) = v_{0x} + \frac{F_x}{m}t \quad x(t) = x_0 + v_{0x}t + \frac{1}{2}\frac{F_x}{m}t^2$$

Varying net force

- The net force on an object can change magnitude and direction:
 - As a function of time.
 - As a function of position.
 - As a function of velocity.
- Then, the acceleration is not constant, and we have to start with:

$$\frac{d\vec{v}}{dt} = \vec{a} = \frac{\vec{F}^{net}}{m}$$
$$\frac{d\vec{r}}{dt} = \vec{v}$$

and integrate...

Linear motion, force varies with time

- If the magnitude of the force changes with time:

$$\vec{a}(t) = \frac{\vec{F}^{net}(t)}{m}$$

- The kinematics equations are “straightforward”:

$$\frac{d\vec{v}}{dt} = \frac{1}{m}\vec{F}^{net}(t)$$

$$\therefore \vec{v}(t) = \vec{v}_0 + \frac{1}{m} \int_0^t \vec{F}^{net}(t) dt$$

$$\therefore \vec{r}(t) = \vec{r}_0 + \int_0^t \vec{v}(t) dt$$

just some integrals that you can look up if you know the function $\vec{F}^{net}(t)$...

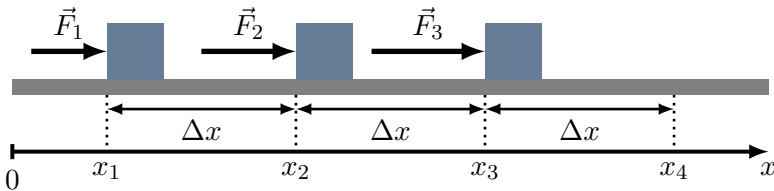
Linear motion, force varies with position

- If the magnitude of the force changes with position of the object:

$$a_x(x) = \frac{F_x^{net}(x)}{m}$$

- We cannot “simply integrate” the equations that define acceleration from velocity and velocity from position:

$$\frac{dv_x}{dt} = a_x(x) \rightarrow v_x(t) = v_{0x} + \int_0^t a_x(x) dt$$



A block slides along x with a force that changes depending on position.

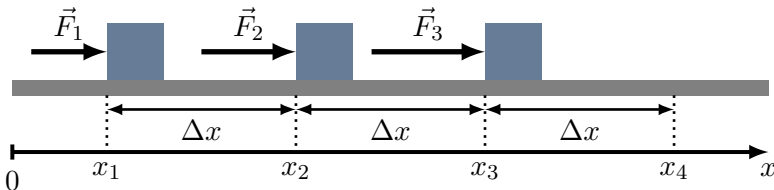
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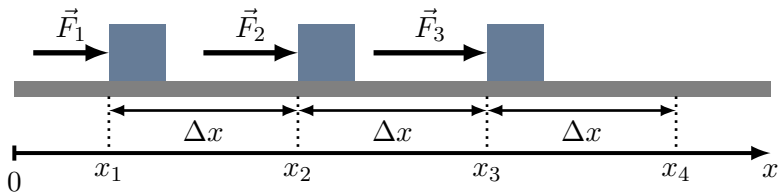
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A block slides along x with a force that changes depending on position.

Breaking it up

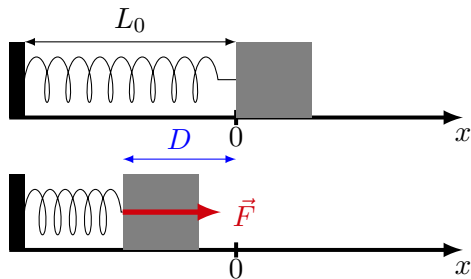
- When things vary continuously, we can break up the motion into small segments where things are constant (in this case, things = force, acceleration).



A block slides along x with a force that changes depending on position.

Applying this to a block on a spring

- A block is pushed against a spring, compressing it by a distance D .
- The force on the block (exerted by the spring) changes with position.

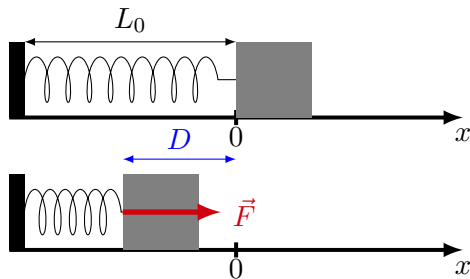


Compressing a spring by a distance D .

Applying this to a block on a spring

- A block is pushed against a spring, compressing it by a distance D .
- The force on the block (exerted by the spring) changes with position.
- The velocity (squared) of the block as a function of position is given by:

$$v^2(x) = v_0^2 + \frac{2}{m} \int F(x) dx$$



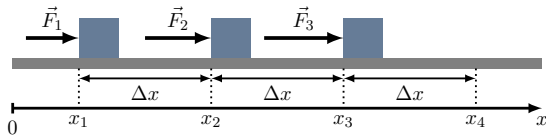
Compressing a spring by a distance D .

Force varying with position (recap)

- We can find the speed squared by applying the kinematic equation multiple times:

$$v_1^2 = v_0^2 + 2\frac{F_1}{m}\Delta x$$

$$v_n^2 = v_0^2 + \frac{2}{m} \sum F_i \Delta x$$



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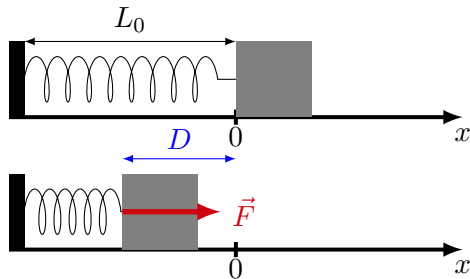
Force varying with position (recap)

- A block is pushed against a spring, compressing it by a distance D . The force on the block (exerted by the spring) changes with position:

$$F(x) = -kx$$

- Having released the block at rest ($v_0 = 0$) from $x = -D$, its speed when it reaches $x = 0$ is:

$$\begin{aligned} v^2(x=0) &= -\frac{2}{m} \int_{-D}^0 kx dx = -\frac{2}{m} \left[\frac{1}{2} kx^2 \right]_{-D}^0 \\ &= \frac{k}{m} D^2 \end{aligned}$$



Compressing a spring by a distance D .

Mathematical Applications

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Mathematically, more formally

- We know acceleration as a function of position:

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- We can thus write

$$dv = a(x)dt = \frac{a(x)}{v}dx$$

Mathematically, more formally

- Think of the Chain Rule for the same result:

$$\frac{dv}{dx} = \frac{dv}{dt} \frac{dt}{dx} = a \frac{1}{v}$$
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- We can “add together” things on the right and the left, and they must still be equal:

$$\int_{v_0}^{v(x_1)} v dv = \int_{x_0}^{x_1} a(x) dx$$
$$\frac{1}{2} v^2(x) - \frac{1}{2} v_0^2 = \int_{x_0}^{x_1} a(x) dx$$
$$\therefore v^2(x) = v_0^2 + 2 \int_{x_0}^{x_1} a(x) dx = v_0^2 + \frac{2}{m} \int_{x_0}^{x_1} F(x) dx$$

Mathematically, more formally

- With constant acceleration, this just gives the kinematics equation:

$$\begin{aligned}v^2(x_1) &= v_0^2 + 2 \int_{x_0}^{x_1} a(x) dx \\&= v_0^2 + 2a(x_1 - x_0)\end{aligned}$$

- This is also where kinetic energy comes from (next week!)

$$\begin{aligned}v^2(x_1) &= v_0^2 + \frac{2}{m} \int_{x_0}^x F(x) dx \\ \therefore \frac{1}{2}mv^2(x_1) - \frac{1}{2}mv_0^2 &= \int_{x_0}^x F(x) dx\end{aligned}$$

Final kinetic energy minus initial kinetic energy is equal to Work done...

Differential equations

- This is a differential equation:

$$v dv = a(x) dx$$

- We could write it as:

$$\frac{dv}{dx} = \frac{a(x)}{v}$$

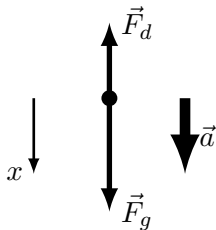
- The derivative of a function $\frac{dv}{dx}$ depends on the function $(v(x))$ itself.
- It is a **separable differential** equation, because we can separate all that depends on v and x onto separate sides of the equation (as in the first line). This allows us to find $v(x)$ by integrating both sides, as in the last slides.
- Separable = usually easy to solve.
- At this point, **you're not expected to solve a differential equation**, but you should understand the concept and how we're solving a separable equation.

Force varying with velocity (e.g. drag)

- The force of drag (aka air resistance) depends on velocity.
- Ultimately, it depends on the actual situation, sometimes it depends on v , sometimes on v^2 .
- The force also depends on the medium through which the object moves, and the size of the object.
- Can model this as:

$$\vec{F}_d = -b\vec{v}$$

$$\vec{F}_d = -bv^2\hat{v}$$

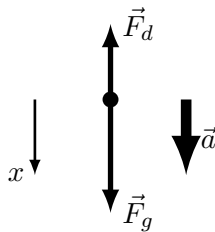


A model for drag.

Example: Drag linear with velocity

- We assume drag is linear (and write the x component):

$$F_d = -bv$$



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Example: Drag linear with velocity

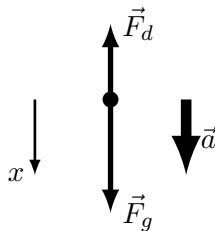
- We assume drag is linear (and write the x component):

$$F_d = -bv$$

- Writing Newton's Second Law (which is a differential equation):

$$\sum F - x = mg - bv = ma$$

$$mg - bv = m \frac{dv}{dt}$$



A model for drag.

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$$F_d = -bv$$

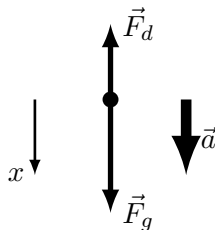
- Writing Newton's Second Law (which is a differential equation):

$$\sum F - x = mg - bv = ma$$

$$mg - bv = m \frac{dv}{dt}$$

- It's separable:

$$\frac{dv}{mg - bv} = -\frac{b}{m} dt$$

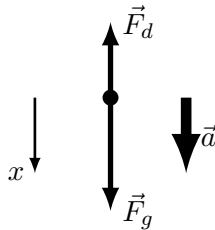


A model for drag.

Example: Drag linear with velocity

- So we can integrate both sides:

$$\int_0^{v(t_1)} \frac{dv}{mg - bv} = - \int_0^{t_1} \frac{b}{m} dt$$



A model for drag.

Example: Drag linear with velocity

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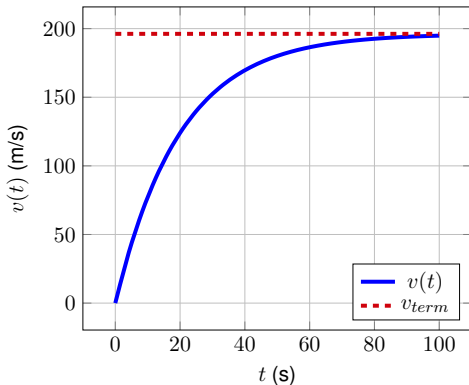
$$\int_0^{v(t_1)} \frac{dv}{mg - bv} = - \int_0^{t_1} \frac{b}{m} dt$$

- to find (after some math):

$$v(t_1) = \frac{mg}{b} \left(1 - e^{-\frac{b}{m}t_1} \right)$$

The speed increases asymptotically to the “terminal velocity”. In the limit $t \rightarrow \infty$, we have:

$$v(t \rightarrow \infty) = \frac{mg}{b}$$



Speed as a function of time for an object that started from rest and falls with drag. The object has a mass of $m = 10$ kg and the coefficient of drag is given by $b = 0.5 \text{ N} \cdot \text{s/m}$.

Building Models Using Newton's Laws

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Reminder: Applying Newton's Laws

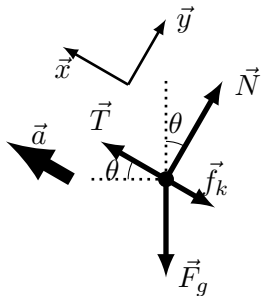
Strongly recommend the following procedure for model building:

1. **Make a very good Free Body Diagram** for each mass that shows:
 - 1.1 All the forces ("what could be pushing or pulling on the mass?").
 - 1.2 The acceleration vector.
 - 1.3 Relevant angles.
 - 1.4 Your choice of coordinate system (make $+x$ parallel to acceleration).
2. Read off the free-body diagram(s) to write Newton's Second Law, you get one equation per dimension:

$$\sum F_x = ma \quad \sum F_y = 0$$

(if you chose x-axis parallel to acceleration, not always possible).

3. You now have a series of equations, just do some math to solve for any variables you need.



Reading off your very good free body diagram

Reading off the diagram to write Newton's Second Law:

$$\sum F_x : \quad T - f_k - F_g \sin \theta = ma$$

$$\sum F_y : \quad N - F_g \cos \theta = 0$$

Newton's Laws and Circular motion

- In circular motion (motion along a circle):
 - The velocity is always tangent to the circle.
 - **There is always a centripetal component of acceleration** (radial acceleration, perpendicular to the velocity, responsible for changing the direction of $\vec{v}$).
 - There can be a tangential component of the acceleration, if the object changes speed (**non-uniform** circular motion).
 - The angular quantities (θ, ω, α) are related to the tangential quantities (e.g. $v = \omega R$, $a_T = \alpha R$).
- The sum of the forces must be parallel to the total acceleration:

$$\sum \vec{F} = m\vec{a}$$

- Can break this up into radial and tangential components (a good choice of xy):

$$\sum F_R = ma_R = m\frac{v^2}{R} = m\omega^2 R \quad (\text{Radial component, never zero})$$

$$\sum F_T = ma_T = m\alpha R \quad (\text{Tangential component, zero if uniform circ. mot.})$$

Two masses on accelerating turntable

- Both tangential and radial acceleration, only force of static friction can result in acceleration.
- Both masses have the same angular velocity (ω) and angular acceleration (α).
- Masses stay while $f_s \leq \mu_s N = \mu_s mg$.
- Tangential (linear) acceleration (constant):

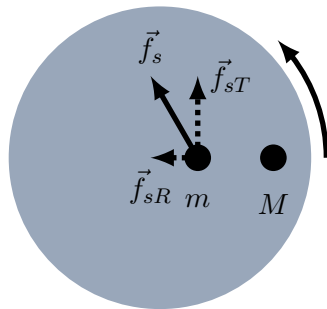
$$ma_T = m\alpha r = f_{sT}$$

- Radial (linear) acceleration (increasing):

$$ma_R = m\omega^2 r = f_{sR}$$

- Magnitude of the force of friction:

$$f_s = \sqrt{f_{sT}^2 + f_{sR}^2} = mr\sqrt{\alpha^2 + \omega^2} \leq mg$$
$$\therefore r\sqrt{\alpha^2 + \omega^2} \leq g$$



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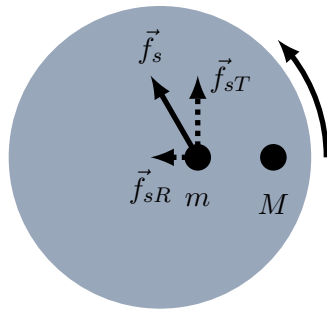
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Since α and ω are the same for both masses, the one with the larger value of r will fall off first, regardless of mass!

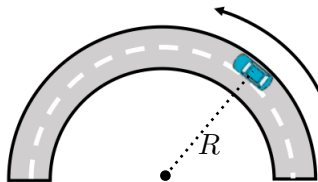
Modeling *uniform* circular motion

- Uniform means constant speed (no tangential acceleration).
- The **net acceleration** is towards the centre of the circle:

$$a_C = a_R = m \frac{v^2}{R}$$

- The **sum of the forces** is directed towards the centre of the circle:

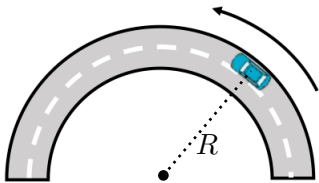
$$\sum F_R = m \frac{v^2}{R}$$
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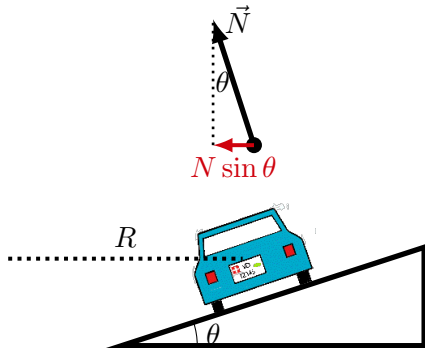
Top view of car going around a corner

Banked curves

- By banking the road, one can change the direction of the normal force so that it, too, can have a component towards the centre of the circle



Top view of car going around a corner

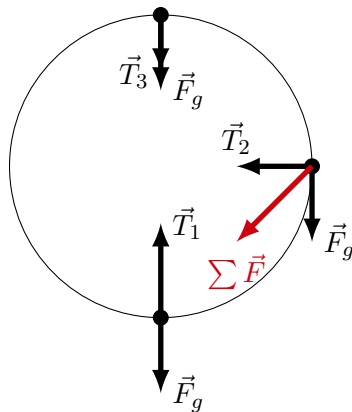


Side view of car going around a banked curve.

Motion in a vertical circle

- What you did as an experiment in RA5.
- A ball swinging around a rope, with a force of tension.
- A roller coaster, with a normal force instead of tension.

It can never be uniform circular motion, there will always be a tangential component to acceleration from gravity (except at the top and bottom).



A mass on a string going around in a vertical circle. The sum of the forces only points to the centre of the circle at the top and bottom.